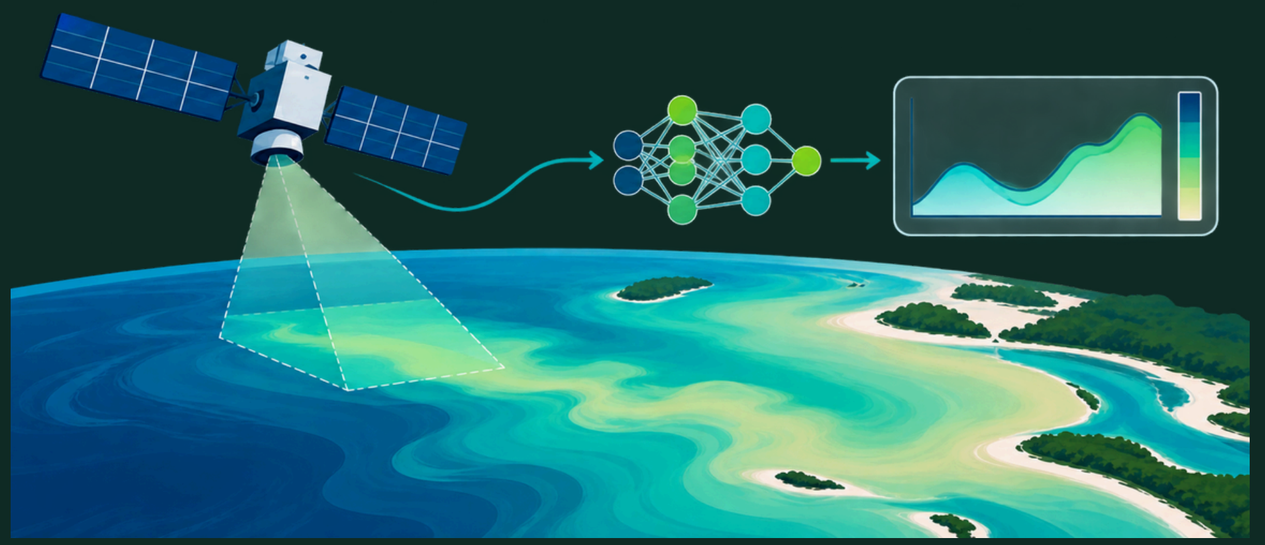
2107541 Machine Learning for Environmental Applications

Monitoring Water Quality from Space: Machine Learning for Satellite-Based Turbidity Estimation



Final Project Part II AY2026 S1

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Outlines

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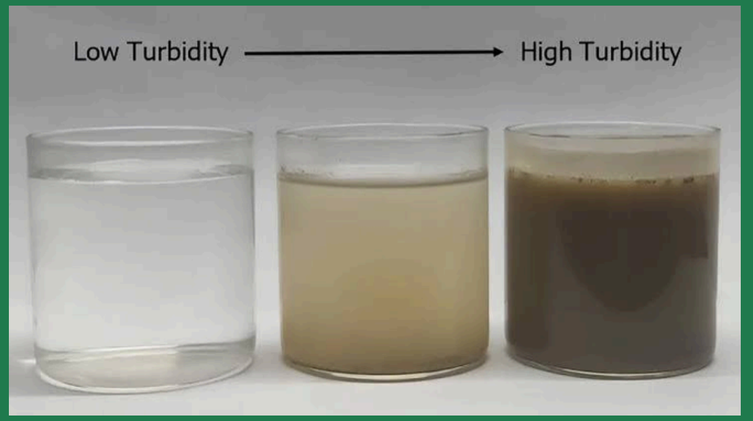
00 | Introduction

Purpose/Target



--> Predict Water Turbidity (TSS) at Large Scale <--

“From point-based water sampling to area-wide turbidity mapping”



What is Water Turbidity?

- **Meaning** : Water having suspended particles (i.e. sediment , silt)
- **Existence** : Lake , Plankton , Natural water currents
- **Affects**
  - Stress on fish and habitats
  - Higher cost for drinking-water treatment
- **Measurement**
  - Turbidity (How can light scatter in water) in **NTU** unit
  - Concentration of Total Suspended Solids (**TSS**) in **mg/L** unit



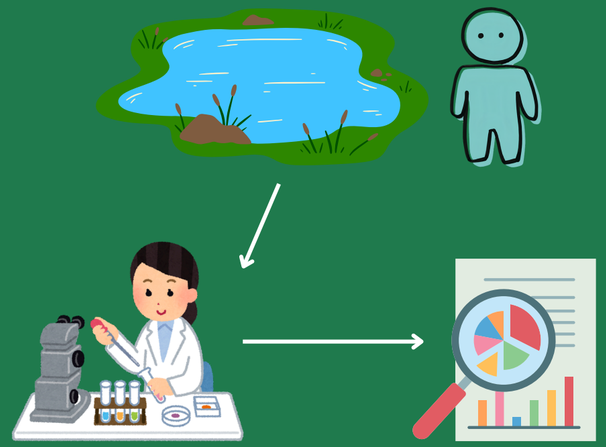
01 | Environmental Problems

Key Point : Unknown Overall Water Quality

Old Procedure

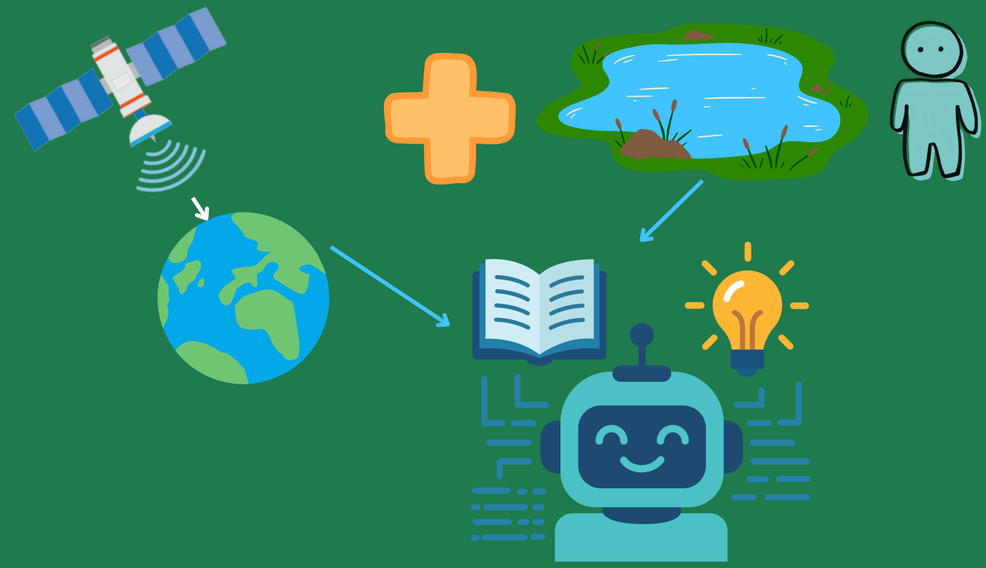
Water collected from natural resources i.e. Lake then sent to testing Lab

Problems : Slow & Expensive & Inaccurate



New Approach

Using Satellites to cover whole water body frequenty together with filed samples to train & check model



New Measurement

Concentration of Total Suspended Solids (mg/L)

O2 | Why Machine Learning?

Key Point : Linear relationships aren’t enough for predicting turbidity

Q1 : Why Linear Relationships isn’t enough to predict?

A : Two reasons are concluded

1. “Water Saturation Value” isn’t in linear relationship , depends on this equation below

$$Reflectance(R) \propto \frac{Backscattering(b_b)}{Absorption(a) + Backscattering(b_b)}$$

This table below shows how TSS changed over on light reflected (Assume as sedimented +10 mg/L)

TSS (mg/L)	10 –> 20	50 → 100	200 → 210
Increase in reflected red light	Large	Moderate	Almost None

2. Water have many factors that we can’t only judge water turbidity as only factor

Factors affected

TSS Concentration | Chrolophyll | CDOM | Atmosphere | Benthic Substrate

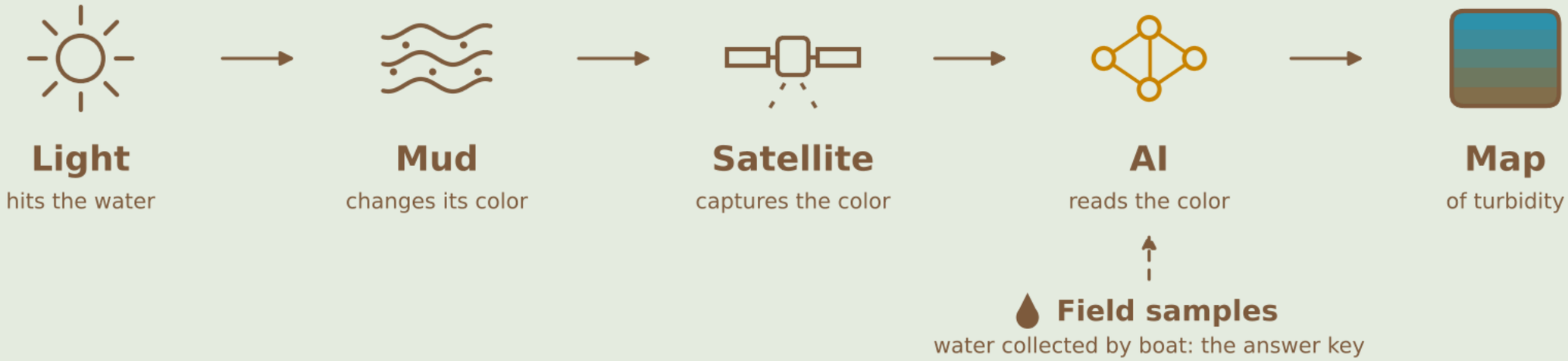
O2 | Why Machine Learning?

Key Point : Linear relationships aren’t enough for predicting turbidity

Q2 : Why use satellites to collect data?

A : Satellites see the whole water body, again and again, and water color reveals what is in it.

- **Coverage**: One image covers the entire lake, and each pixel is a measurement point (a boat samples one spot at a time)
- **Frequency**: They revisit on their own, e.g. Sentinel-2 about every 5 days in clear weather
- **Signal**: Sediment changes water color: clear water looks dark blue, muddy water looks bright brown



03 | Machine Learning Model / Framework

What Should you know before model? → Columns Meaning

Group 1 : Satellites (Landset 7) Surface Reflectance



blue	green	red
230	243	215

nir	swir1	swir2
383	228	172

Note : These values need to be multiply with
scale_factor = 1e-4 before use
or multiply with 100 to get surface_reflectance_percentage

Feature to calculate Satellite Index

NDTI

Measure soil tillage residues or water cloudiness (turbidity)

$$NDTI = \frac{Red - Green}{Red + Green}$$

NDWI

Detect open surface water and measure moisture in plants

$$NDWI = \frac{Green - NIR}{Green + NIR}$$

NDSSI

measures suspended sediment and turbidity in water bodies using satellite imagery.

$$NDWI = \frac{Blue - NIR}{Blue + NIR}$$

Group 2 : Image Quality



clouds	17	Percentage of cloud cover in scene [%]
p_water	97	Percentage of water in surrounding pixels [%]
qa	68	Landsat pixel quality code
qa_sd	0.996	Variance of qa around pixels
time_diff	0.996	Time Gap between satellites and water samples [Hours]
pixelCount	138	Average number of pixels around sample point [pixels]

Group 3 : Variances around point

Represents Standard Deviation of each bands

blue_sd	green_sd	red_sd	nir_sd	swir1_sd	swir2_sd
25.9	22.8	25.7	30.7	23.2	29.0

Note : These values need to be multiply with
scale_factor = 1e-4 before use
or multiply with 100 to get surface_reflectance_percentage

O3 | Machine Learning Model / Framework

What Should you know before model? → Columns Meaning

Group 4 : Field Samples



Note : These features won't be use to train in model

tss	NA	(Not Measured) Concentration of Total Suspended Solids [mg/L]
chl_a	3.1	Chlorophyll-a (algae) [µg/L]
doc	7.99	Dissolved organic carbon [mg/L]
secchi	3.83	Depth at which a white disk disappears (Depper = Clearer) [metre]
tis	NA	Total Inorganic Solids [mg/L]
p_sand	NA	Percentage of sands in water [%]

Group 5 : Stations Info



SiteID	8348	Sampling station ID. Used as the group in GroupKFold
lat , long	45.508, -67.829	Station coordinates (north positive, west negative)
.geo	{"Point", [-67.83, 45.51]}	Station coordinates in GeoJSON format
type	Lake	Type of water body
source	LAGOS	Database the field sample came from
date , date_only	2004-08-23, TRUE	date : Sampling date date_only : Status to check if format is date or not
TZID	America/New_York	Local time zone of the site
date_unity	2004-08-23T12:00Z	Probably a placeholder time (noon)
date_utc	2004-08-23T16:00Z	Sampling time converted to UTC
time	2004-08-23T15:02:20Z	Satellite overpass time
endtime	00:00:00	(Uncleared Notation)

03 | Machine Learning Model / Framework

What Should you know before model? → Columns Meaning

Group 6 : Satellites Background Info



tss	7	Landsat satellite number that took the image
path, row	10 , 28	Position of the scene in the Landsat grid
landsat_id	LE07_L1TP_010028_20040823_20160926_01_T1	

Details for landsat_id

LE07 → Landsat 7 (ETM+ sensor)
L1TP → processing level
010028 → path 010, row 028
20040823 → image date: 23 Aug 2004
20160926 → processing date
01 → Collection 1
T1 → Tier 1 (best quality)

Additional Resources

Full harmonized in-situ datasets

https://doi.org/10.6084/m9.figshare.8139362

Cite | Download (196.88 MB) | Share | Embed | + Collect

Dataset posted on 2019-10-11, 04:33 authored by Matthew Ross

There are two files in this dataset, both of which are part of the AquaSat dataset (paper DOI HERE) and code here: https://github.com/GlobalHydrologyLab/watersat

The first (wqp_lagos_unity.csv) is a dataset containing all merged observations in the LAGOS-NE databse (https://doi.org/10.1093/gigascience/gix101) and data from the Water Quality Portal. This includes concentration data for chlorophyll a (in ug/L), total suspended solids (in mg/L), dissolved organic carbon (in mg/L), and secchi disk depth (in m). There is also SiteID names for each observation.

Landsat SR-Derived Spectral Indices Pixel Quality Band

By Landsat Missions

HOME
LANDSAT MISSIONS
OPERATING THE LANDSAT MISSIONS
SCIENCE
PRODUCT INFORMATION
DATA
GLOSSARY AND ACRONYMS
MULTIMEDIA
PUBLICATIONS
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Return to Landsat Surface Reflectance-Derived Spectral Indices Overview

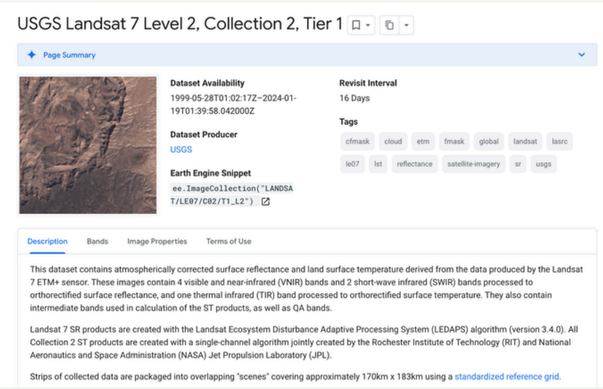
A Pixel Quality Assurance (pixel_qa) band is provided with all Landsat Surface Reflectance-derived Spectral Indices.

Each pixel_qa band is provided for Landsat 4-5, Landsat 7 and Landsat 8 data. The band is in unsigned 16-bit format, whose values are bit-packed and provide information pertaining to a pixel condition of fill, clear, water, cloud shadow, snow, cloud (yes/no), cloud confidence and cirrus cloud confidence (Landsat 8 only.)

The tables below display the bits, values, and interpretations for the Pixel Quality Assurance band for Landsat Surface Reflectance-Derived Spectral Indices.

Landsat Surface Reflectance-Derived Spectral Indices Pixel Quality Band: Landsat 4-5, Landsat 7

Bit	Value	Cumulative Sum	Interpretation
0	1	1	Fill
1	2	3	Clear
2	4	7	Water



03 | Machine Learning Model / Framework

ML Flow

Input	Features	Target	Validation	Explanation
6 Landsat bands (scaled x 1e-4)	16 features: bands, indices, ratios, season	TSS [mg/L] log(1 + TSS)	Split by SiteID R^2, RMSE, MAE	Feature importance SHAP

Model Used

	Baseline	Model #2	Model #3	Model #4
	Linear Regression	Random Forest	XGBoost	Multi-Layer Perceptron (Neural Network)
Category	Linear Regression	Ensemble (Bagging)	Ensemble (Boosting)	Neural Network
Feature	Same 16 features for all 4 models blue, green, red, nir, swir1, swir2 · NDTI, NDWI, NDSSI · R/G, R/B, G/B, NIR/R · log(R+G+B) · sin/cos(day of year)			
Hyperparameters	– (StandardScaler only)	n_estimators = 200 min_samples_leaf = 5 max_features = 0.33 max_samples = 0.5	n_estimators = 300 max_depth = 4 learning_rate = 0.05 subsample = 0.8 colsample_bytree = 0.8	layers (64, 32), ReLU Adam, lr = 0.001 batch = 256, alpha = 1e-4 early stopping
Explanation	Coefficients	Feature importance	SHAP (beeswarm) + feature importance	Scores only (no built-in importance)
Reason to use	Basic procedure (baseline)	Does curving the fit help the data?	Does error-correcting help the data?	Can a neural net learn a better curve?

03 | Machine Learning Model / Framework

Data Preparation → 603,432 raw matchups become 128,603 clean samples

Cleaning steps (rows kept)

Raw AquaSat matchups
all rows in the file



603,432

1 · Has a TSS value (> 0 mg/L)
removed 443,613



159,819

2 · Valid reflectance (0 – 10,000)
removed 2,399



157,420

3 · Clear-water pixel (QA 68 / 324)
removed 21,527



135,893

4 · Mostly water (p_water >= 50%)
removed 7,290



128,603

Missing TSS is the big loss: only 1 in 4 matchups has a TSS measurement.

Target: log(1 + TSS)

TSS is extremely right-skewed (median 11 mg/L, max 73,900 mg/L). Log scale cuts skewness from 129 to 0.83, so a few storm samples do not dominate.

Split by station, not by row

Each station is sampled many times, so a random split leaks. GroupShuffleSplit 80/20 by SiteID: train 102,142 rows (16,903 stations), test 26,461 rows (4,226 unseen stations). CV: GroupKFold, 5 folds.

16 input features

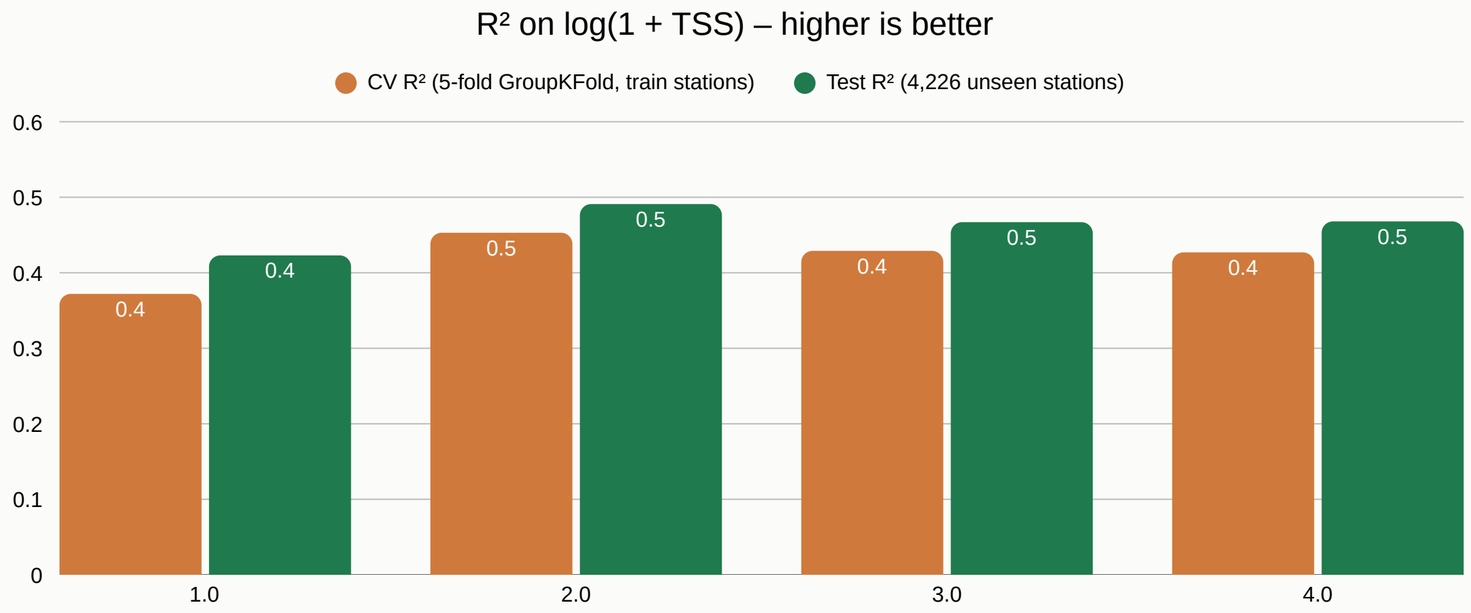
6 bands (x 1e-4) · NDTI, NDWI, NDSSI · R/G, R/B, G/B, NIR/R · log(R+G+B) · season sin/cos(day of year)

04 | Real Application / Research Findings

Key Point : Satellite-based water quality monitoring is already used in practice, and our experiment shows how much ML adds.

Part B: Our experiment (4 ML models)

- Non-linear models beat the linear baseline: test R^2 0.423 → 0.47–0.49
- Random Forest is the best model before tuning (test R^2 0.491)
- Every score is measured on 4,226 stations the models never saw



Test set	Linear Reg.	Random Forest	XGBoost	MLP
RMSE (lower = better)	0.862	**0.809**	0.828	0.827
MAE (lower = better)	0.646	**0.598**	0.618	0.621

04 | Real Application / Research Findings

Key Point : Satellite-based water quality monitoring is already used in practice, and our experiment shows how much ML adds.

Part C: Hyperparameter tuning

- Both models chose more complexity: deeper trees, more trees
- Train R^2 rises more than CV R^2, but CV still improves

Random Forest

0.453 → **0.465**

CV R^2 before → after tuning
Train R^2 0.656 → 0.805
Best: min_samples_leaf = 1, max_features = 0.5, max_depth = None

XGBoost

0.429 → **0.464**

CV R^2 before → after tuning
Train R^2 0.462 → 0.683
Best: n_estimators = 800, max_depth = 8, learning_rate = 0.03

Method: RandomizedSearchCV (4 random combinations) scored with GroupKFold (3 folds by SiteID); the test set stays untouched.
Gain is only +0.01 to +0.04 R^2 → the features, not the settings, limit the accuracy.
Linear Regression has no hyperparameters; the MLP was not tuned (too slow).

04 | Real Application / Research Findings

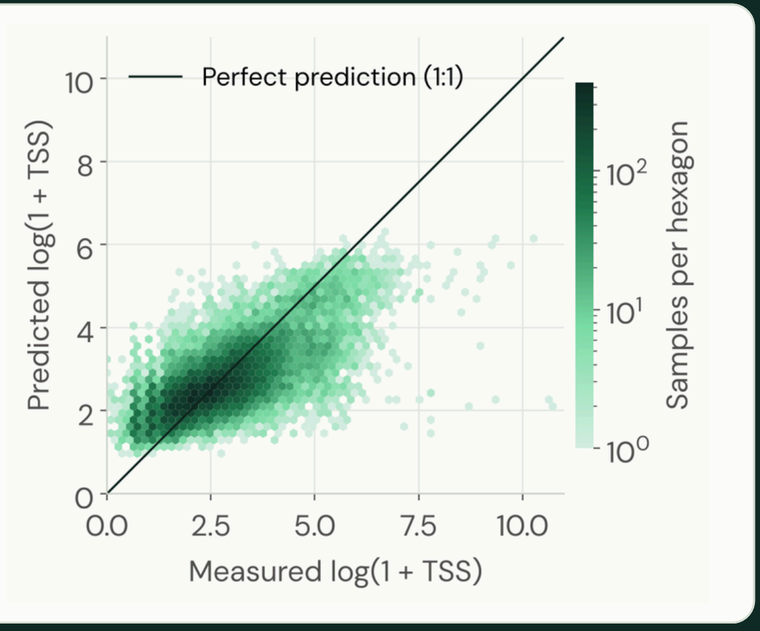
Key Point : Satellite-based water quality monitoring is already used in practice, and our experiment shows how much ML adds.

Part D: Best model (tuned Random Forest)

0.503
R^2 score (higher = better)
half of the variation, unseen stations

0.800
RMSE (lower = better)
on log(1 + TSS), test set used once

69%
within 2x of lab TSS
between half and double the lab value



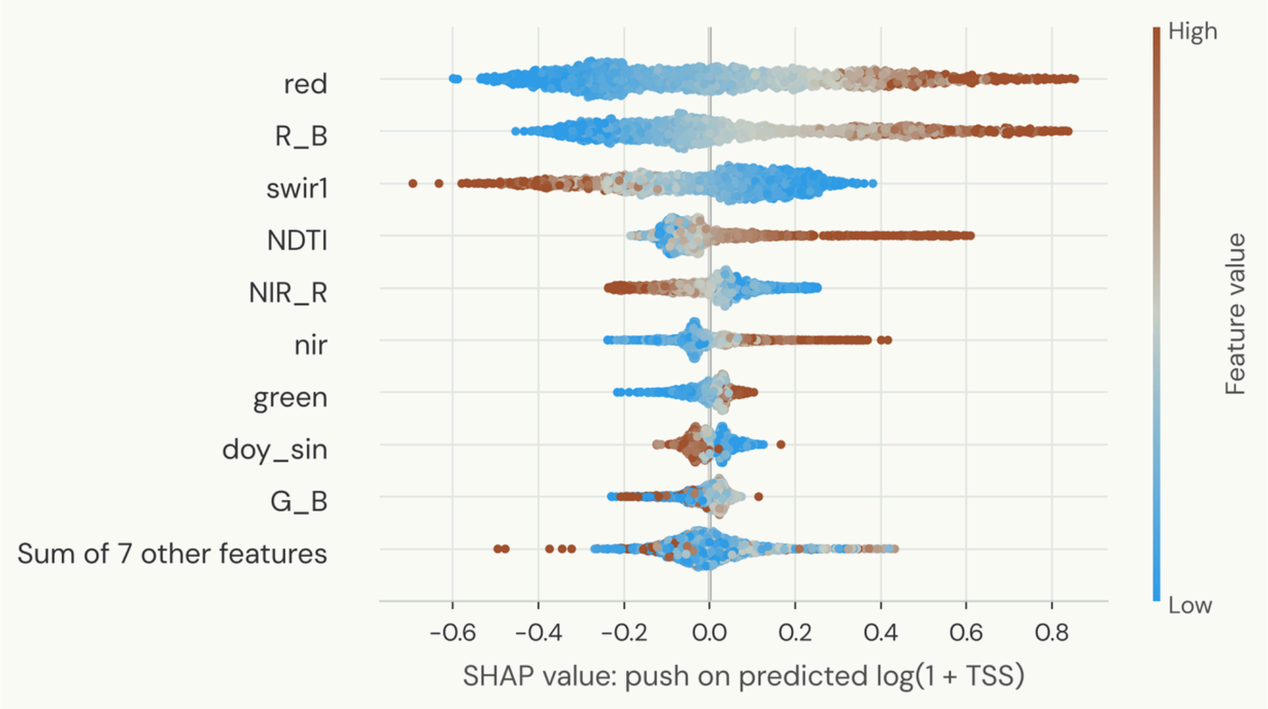
Predicted vs. actual (test stations)

- Each hexagon holds test samples; darker = more samples. The black line is a perfect prediction
- The cloud follows the line → the model ranks clear vs. turbid water correctly
- The cloud is flatter than the line → clear water is over-predicted, very turbid water under-predicted
- MAE 0.586 on the log scale → predictions are off by about 1.8x on average

05 | Environmental Meaning / Interpretation

Key Point : The model follows water physics – reliable for relative, not exact, turbidity

What pushes the prediction up or down? (SHAP, XGBoost)



Each dot = one test sample. Brown = high feature value, blue = low. Right of 0 = higher predicted TSS.

More red light → more TSS

Sediment scatters red light back to the satellite: more backscattering, more reflectance (equation in section 02). Red is the top feature in SHAP and in RF / XGBoost importance.

Higher red/blue & NDTI → more TSS

Muddy water shifts from blue-green to brown. Ratios also cancel haze and sun-angle effects that brighten every band.

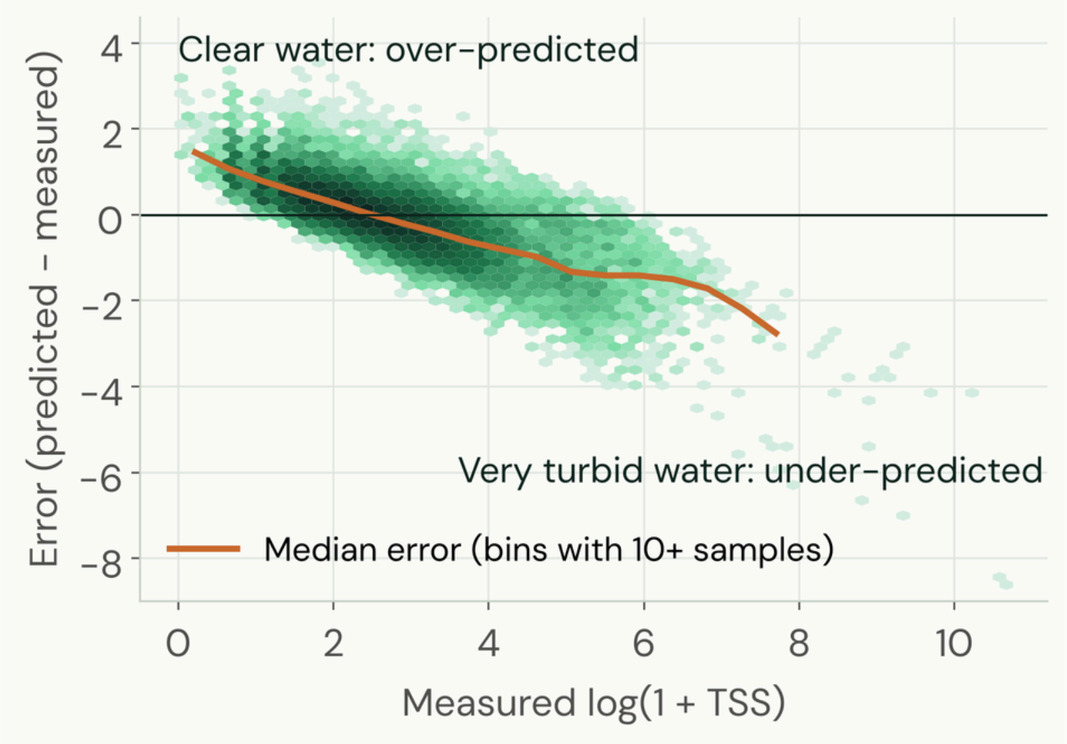
More SWIR → less TSS

Water absorbs SWIR, so SWIR signal over water mostly means glint, haze or nearby land. The model likely uses it to discount false brightness.

05 | Environmental Meaning / Interpretation

Key Point : The model follows water physics – reliable for relative, not exact, turbidity

Where does the best model go wrong? (test stations)



Good for

Mapping relative turbidity and screening thousands of unsampled lakes and rivers. Turbidity class right for 67% of samples (vs. 45% by guessing); clear water is almost never called highly turbid (0.4%).

Not for

Exact mg/L for permits and compliance – average error is about 1.8x, so field samples are still needed.

Weakest where it matters

Storm and flood plumes (200–1,000 mg/L) are predicted about 4x too low: red reflectance saturates at high TSS (section 02:
200 → 210 mg/L gives almost no change).

06 | Room for Improvement

Key Point : Better inputs, not bigger models, will move accuracy

Limitations

- Only satellite colour is used – no rainfall, land use or water depth
- Range is compressed: clear water over-predicted, turbid water under-predicted
- Streams (49%), estuaries (27%) and lakes (23%) are mixed but not labelled; Landsat 5 / 7 / 8 differences ignored
- Small tuning budget (4 combinations); MLP not tuned
- Image can be up to ±1 day from the sample; 30 m pixels miss narrow rivers

Next steps

- Add water-body type (stream / estuary / lake) and Landsat sensor (5 / 7 / 8) as input features
- Clip extreme band ratios (red/blue reaches 41 on very dark pixels)
- Add drivers of sediment: recent rainfall, watershed land use, depth
- Weight rare high-TSS samples; wider search; validate by whole watershed
- Sentinel-2 (10 m, ~5-day revisit) for small rivers and ponds

Extra : Repository & Web Visualizer

Repository

Web Visualizer

References

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